Correspondence between collections

Two perfect condensations of the same observables satisfy

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) = H(Z_{\supseteq A} \mid Y_{\supseteq A}) = 0.$$

Here $Y_{\supseteq A} = (Y_C : A \subseteq C)$ is the latent at A together with those at all of its supersets. The corresponding collections determine one another almost surely.

The proof has three steps: combine two witnesses, transfer through observations, and intersect contributing families.

A common joint law

The two models specify laws for (X, Y) and (X, Z) with the same X marginal. Choose a joint law for (X, Y, Z) preserving both.

One possible choice is

$$p(x, y, z) = p(x)\, p(y \mid x)\, p(z \mid x).$$

The theorem holds for any such coupling.

The quantitative theorem

Fix nonempty $A \subseteq I$ and let $k = |A|$. For latent models Y, Z of the same observables,

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) \leq \sum_{i \in A} H(Y_{\supseteq A} \mid X_i) + (k-1)\eta_Z.$$

Here η_Z bounds conditional dependence between upward-closed families given their intersection. If every reconstruction defect and η_Z are at most ε, the bound reads

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) \leq (2k-1)\varepsilon.$$

This direction uses recovery of the tower from the observations and the Markov defect of Z.

The two-witness lemma

For finite-valued U, M, P, Q,

$$H(U \mid M) \leq H(U \mid M, P) + H(U \mid M, Q) + I(P; Q \mid M).$$

If both witnesses (M, P) and (M, Q) recover U, and their remaining information is conditionally independent given M, then M recovers U.

The inequality also tracks errors when these requirements hold approximately.

The entropy identity behind the lemma

Subtract the left side from the right side:

$$\begin{aligned} H(U \mid M,P) + H(U \mid M,Q) + I(P;Q \mid M) \\ - H(U \mid M) \\ = I(P;Q \mid U,M) + H(U \mid M,P,Q) \;\geq\; 0. \end{aligned}$$

This follows from the chain rule,

$$H(U,V \mid W) = H(U \mid W) + H(V \mid U,W),$$

and nonnegativity.

Transfer through an observation

The contributing family $\supseteq\{i\}$ satisfies

$$H(X_i \mid Z_{\supseteq\{i\}}) = 0.$$

Therefore, for any target U,

$$H(U \mid Z_{\supseteq\{i\}}) \leq H(U \mid X_i).$$

The contributing latents determine X_i, so they provide at least its information about U.

Intersecting the contributing families

For $A = \{i_1, \ldots, i_k\}$, set

$$\mathcal{G}_j = \bigcap_{\ell=1}^{j} \supseteq\{i_\ell\}.$$

An address C survives the final intersection exactly when it contains every element of A:

$$\mathcal{G}_k = \{C : A \subseteq C\} = \supseteq A.$$

Each $\supseteq\{i\}$ is upward closed. Intersections preserve upward closure.

The smaller index carries the bigger tuple

Each intersection adds a requirement, so the families shrink:

$$\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \cdots \supseteq \mathcal{G}_k = \supseteq A.$$

An address survives step j only if it already survived step $j-1$.

A family with more addresses carries a longer tuple of latents, so the tuples run the other way: every latent in $Z_{\mathcal{G}_j}$ also appears in $Z_{\mathcal{G}_{j-1}}$, and also in $Z_{\supseteq\{i_j\}}$, since $\mathcal{G}_j = \mathcal{G}_{j-1} \cap \supseteq\{i_j\}$.

One intersection step

Use the two-witness lemma with

$$M = Z_{\mathcal{G}_j}, \qquad P = Z_{\mathcal{G}_{j-1}}, \qquad Q = Z_{\supseteq\{i_j\}}.$$

The tuple M is already contained in both P and Q. Hence

$$H(U \mid Z_{\mathcal{G}_j}) \leq H(U \mid Z_{\mathcal{G}_{j-1}}) + H(U \mid Z_{\supseteq\{i_j\}}) + I(Z_{\mathcal{G}_{j-1}}; Z_{\supseteq\{i_j\}} \mid Z_{\mathcal{G}_j}).$$

Both families are upward closed and $\mathcal{G}_j$ is their intersection, so the last term is at most η_Z.

Counting the error terms

Iterating the previous step down from $\mathcal{G}_k = \supseteq A$ to $\mathcal{G}_1 = \supseteq\{i_1\}$ uses $k-1$ intersections, and leaves k terms, one per element of A. For any target U,

$$H(U \mid Z_{\supseteq A}) \leq \sum_{i \in A} H(U \mid Z_{\supseteq\{i\}}) + (k-1)\eta_Z.$$

Transfer through an observation bounds each of those k terms by $H(U \mid X_i)$. Taking the tower $Y_{\supseteq A}$ as the target turns them into the reconstruction defects of Y:

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) \leq \sum_{i \in A} H(Y_{\supseteq A} \mid X_i) + (k-1)\eta_Z.$$

If every reconstruction defect is at most δ and $\eta_Z \leq \eta$, the right side is $k\delta + (k-1)\eta$.

What the bound compares

The reverse direction has the same form:

$$H(Z_{\supseteq A} \mid Y_{\supseteq A}) \leq \sum_{i \in A} H(Z_{\supseteq A} \mid X_i) + (k-1)\eta_Y.$$

For perfect models both sides have zero recovery error and Markov defect, so both conditional entropies vanish. These are bounds on **whole collections**.

Small errors for each individual latent can accumulate across a collection, so the tower hypothesis $H(Y_{\supseteq A} \mid X_i)$ has to be checked directly.

The conditioned score

For a nonempty observation block B, define

$$\mathcal{J}_B = \{A : A \cap B \neq \varnothing\}, \qquad Y_{\supsetneq A} = (Y_C : A \subsetneq C).$$

The score charges each active latent given its strict supersets:

$$\chi(B) = \sum_{A \in \mathcal{J}_B} H(Y_A \mid Y_{\supsetneq A}) \geq H(Y_{\mathcal{J}_B}) \geq H(X_B).$$

The chain rule gives the first inequality; LVM gives the second.

Two terms in the score gap

Let $V_B = \chi(B) - H(Y_{\mathcal{J}_B})$. Then

$$\chi(B) - H(X_B) = V_B + H(Y_{\mathcal{J}_B} \mid X_B).$$

Both terms are nonnegative.

- V_B measures departure from the ordered Markov factorization.
- $H(Y_{\mathcal{J}_B} \mid X_B)$ measures latent information unrecovered from the observations.

More precisely, $V_B = D_{\mathrm{KL}}(P_{Y_{\mathcal{J}_B}} \| Q_B)$, where $Q_B(y) = \prod_{A \in \mathcal{J}_B} P(y_A \mid y_{\supsetneq A})$.

The score ranks models by the queries they serve

Three latent models of the clustering example, all satisfying LVM and the Markov condition. Take B to be a few points inside one cluster.

- One latent carrying the whole tuple: $\chi(B) = H(X)$. Every query pays for all 36 points.
- A chain, with X_j carried at the address $\{j, \dots, 36\}$: $\chi(B) = H(X_1, \dots, X_{\max B})$. Cheap only for early points.
- The hierarchy, with the shape, the group and the cluster centers above the points: writing Θ for the parameters B sits under, $\chi(B) = H(\Theta) + \sum_{i \in B} H(X_i \mid \Theta)$.

Only the hierarchy charges a local query locally, and its gap is $\chi(B) - H(X_B) = H(\Theta \mid X_B)$, the parameter uncertainty a finite block leaves. At $B = I$ the two baselines pay exactly $H(X)$ and the hierarchy pays $H(\Theta \mid X)$ more: the ranking is relative to the blocks actually queried.

Why expect agents to satisfy these conditions?

An agent that answers queries about a block B pays $\chi(B)$, and no latent model does better than $H(X_B)$. The gap splits into exactly our two defects:

$$\chi(B) - H(X_B) = V_B + H(Y_{\mathcal{J}_B} \mid X_B).$$

Short descriptions for the queries an agent cares about *are* small defects. Taking $B = \{i\}$, where $\mathcal{J}_{\{i\}} = \supseteq\{i\}$, gives for $i \in A$

$$H(Y_{\supseteq A} \mid X_i) \leq H(Y_{\supseteq\{i\}} \mid X_i) \leq \chi(\{i\}) - H(X_i),$$

which is exactly the theorem's reconstruction hypothesis.

So the conditions need not be assumed as a coincidence: efficient agents buy them. Efficiency is relative to the blocks an agent actually queries.